

Book chapter for *Psych predicates in Romance languages* (ed. by Kailuweit et al.)

Diathesis alternation and case of Experiencer-object with psych-verbs in Spanish

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Abstract

Psychological predicates pose a challenge to the transitivity framework, as they vary markedly in the assignment of argument structure, both crosslinguistically and intralinguistically. In Spanish, a contrast is found between verbs that take the Experiencer as Subject and the Stimulus as Object (Psych-I class: *amar*, *temer*), and those that take the Experiencer as Object and the Stimulus as Subject (Psych-II: *gustar*, *asustar*). Furthermore, while with some Psych-II verbs, such as *gustar*, the 3rd person clitic Experiencer-Object is systematically marked with the dative case, other verbs, such as *asustar*, show case alternation (accusative / dative) for the Experiencer-Object (*A ella no la/le asustaba el contagio*). Additionally, Psych-II verbs with case alternation in the active voice also allow for a diathesis alternation with the middle voice or “pronominal” construction and the Experiencer as subject (*ella no se asustaba de nada*).

The aim of this study is to explore the factors that motivate the choice between active (Experiencer-Object) and middle (Experiencer-Subject) constructions with Psych-II verbs. The analysis pays particular attention to the choice between accusative and dative case for the object in the active clauses, to determine how case alternation and diathesis alternation are related.

Using data from ADESSE database (265 psych verbs and 5,845 clauses) and CORPES XXI (50 psych verbs and 169,338 clauses), we performed a regression analysis and found that the Experiencer Person, Aspect, and Verb are the main significant factors for the use of the Experiencer-Object vs Experiencer-Subject pattern. Besides, significant predictors for the selection of Accusative vs Dative case of the Experiencer-Object are Zone, Stimulus Person, Aspect and Verb. However, notwithstanding some parallelisms in the predictors, we do not find a significant correlation between diathesis alternation (Subject vs Object Experiencer) and case alternation (Accusative vs Dative Object Experiencer) with verbs that allow for both kinds of alternation. This leads to the conclusion that each construction type has different communicative functions.

Keywords: psychological verbs; case marking; middle voice; diathesis; corpus analysis

1. Introduction

In Spanish, as in other Romance languages, psychological verbs require two participants (Experiencer & Stimulus) and can be construed with 2 opposite syntactic orientations. Some verbs (**Psych-I** verbs such as *amar* ‘love’, *temer* ‘fear’, *sufrir* ‘suffer’, *añorar* ‘miss’ ...) code the Experiencer as Subject and the Stimulus as (Direct) Object, like in (1). Other verbs (**Psych-II** verbs such as *gustar* ‘like’, *desagradar* ‘dislike’, *sorprender* ‘surprise’, *asustar* ‘frighten’ ...) code the Experiencer as (Direct or Indirect) Object and the Stimulus as Subject, like in (2)-(4)

- (1) *Yo no añoro la infancia.* [CIN:043.20]
1SG.NOM NEG miss ART childhood
Exp=Subj Stim=Obj
‘I don’t miss childhood’
- (2) *Me desagrada este Madrid ruidoso* [PAS:043.12]
1SG.AC/DAT dislike this Madrid noisy
Exp=Obj Stim=Subj
‘I dislike this noisy Madrid’
- (3) *No la asustaba el contagio* [CORPES, Arg.]
NEG 3SG.ACC frighten.PSD ART contagio
Exp=Obj Stim=Subj
‘She was not afraid of contagion’
- (4) *A Leila le asustaba casi todo* [CORPES, Arg.]
to Leila 3SG.DAT frighten.PSD almost everything
Exp=Obj Exp Stim=Subj
‘Leila was scared of almost everything’

Within Psych-II verbs we can recognise two subclasses: *Psych-IIa* (= Belletti & Rizzi (1988)’s class II) alternate accusative/dative case in 3rd person clitics [examples (3)-(4) *supra*] and allow for the pronominal construction [= middle voice] with the Experiencer as Subject [example (5)]; whereas *Psych-IIb* (= Belletti & Rizzi’s class III) take exclusively dative 3rd person clitics but not accusative (6), and they do not allow the diathesis alternation (7)

- (5) *Ella no se asustaba (de nada)*
She NEG MV frighten.PSD of anything
Exp=SUBJ PREDMIDDLE Stim=OBL
‘She wasn’t scared (of anything)’
- (6) *Le/*lo/*la desagrada este Madrid ruidoso*
3SG.DAT/*ACC dislike this Madrid noisy
Exp=Obj Stim=Subj
‘(S)he dislikes this noisy Madrid’

- (7) * *Ella no se desagrada (de este Madrid ruidoso)*
 She NEG MV dislike (of this noisy Madrid)

The diathesis alternation exemplified in (3)-(5) is based on a regular semantic correspondence between the object of the active SUBJ-OBJ construction and the subject of the middle-*se* construction. Psych-IIa verbs assign the same semantic role, Experiencer, to the object of (3) and the subject of (5), whereas the Stimulus triggers subject agreement in (3) and takes a preposition or is absent in the middle construction (5). The possibility of this alternation is related to transitivity, given that it is allowed by Psych-IIa verbs, which also allow Accusative case, but not by Psych-IIb verbs.

In the literature, the contrast underlying active / middle-*se* alternation has been related to two semantic dimensions of the event, *causal structure* and *aspectual structure* (Croft 1993; Croft 2012)

Regarding causal structure, even though both alternates deviate from the prototypical model of causation, cause-related differences were found between active and middle-*se* constructions, among which the spontaneity of event and the Experiencer's responsibility and affectedness stand out.

The degree of spontaneity attributed to the event is a factor that has been highlighted in several studies as a basis for choosing between the two structures (cf. Haspelmath 1993, 2016; Melis 1999: 52; Heidinger 2015; García-Miguel 2023: 284). These studies point to the probability of the spontaneous occurrence of the described situation as a relevant factor: the greater the spontaneity, the less the need to express the causal or triggering element of the situation. Thus, spontaneity favours the use of the middle-*se* construction, which does not require an argument that refers to what provokes the emotional process in the Experiencer (i.e. the Stimulus). Even when a cause or Stimulus coded as oblique is mentioned in the clause, the middle-*se* construction implies a defocusing of this participant, since it ‘places the oblique element in the periphery of the event’ (“sitúa en la periferia del evento al elemento oblicuo”; Melis 1999: 55).

The responsibility of the Experiencer in the occurrence of the event has been mentioned by Maldonado (1999: 52), Melis (1999:52), Croft (2012: 235), among others. Maldonado (1999:52) states that the middle indicates an increase in the involvement and responsibility of the Experiencer. Melis (1999: 52) also points out that the subject Experiencer ‘carries

implicitly associated nuances of responsible initiator of the event' (“lleva asociados implícitos matices de responsable iniciador del evento”).

With respect to the affectedness of the Experiencer, Melis (1999) and Vázquez & Miglio (2016) show that the object Experiencer of the active pattern is conceived as more affected than the subject Experiencer of the middle. Usage data reveal a preference for first- and second-person Experiencers to be encoded as objects in the active alternate, in contrast to the third-person Experiencers, who are more often cast as subjects in the middle. This distribution is related to the ‘empathy hierarchy’ (Kuno & Kaburaki 1977, Kuno 1987), which supports the fact that the speaker perceives at first hand (introspectively) how a situation affects him/her (1st person) and, to some extent, how it affects his/her interlocutor (2nd person), whereas it is not so easy to determine how much a third person Experiencer is affected by a mental event.

Regarding aspect, Psych-IIa verbs are assigned a wide variety of interpretations according to the Vendler-Dowty classification (cf. Di Tullio 2004; Vanhoe 2004; Marín 2011; Cifuentes 2015; Melis 2019). Active SUBJ-OBJ constructions have been analysed alternatively as stative or inchoative depending on the case marking of the object Experiencer (dative or accusative) (cf. Ackerman & Moore 1999; Cifuentes 2015), but the limitations of an aspectual classification system based on formal tests, which in many cases give dubious results, have also been pointed out (Kailuweit 2015). Nevertheless, usage-based studies have found significant correlations between perfect tenses and the accusative case, and between imperfect tenses and the dative case (Miglio et al. 2013, Vázquez Rozas & García-Miguel in press). These data can be taken as evidence for the association of the accusative vs dative marking with, respectively, telic vs atelic events.

Differences in aspectual class have also been observed between active and middle clauses with Psych-II verbs. Marín and McNally (2011) identify an inchoative meaning in all middle-*se* constructions, with a further distinction between punctual verbs, such as *enfadarse*, and non-punctual verbs, such as *aburrirse*. Marín (2011), on the other hand, after applying several aspectual combinatorial tests to active constructions, concludes that they behave like stative expressions.

The aim of this study is to investigate the factors that influence the choice between alternating constructions with psychological verbs. Building on the contributions of previous studies, we focus on several linguistic and contextual features that, according to the literature, are involved in (i) the choice between active voice (Exp=Obj) and middle voice (Exp=Subj) of Psych-II verbs, and (ii) the choice between the accusative and dative case for the Exp=Obj of

Psych-II verbs. Finally, the study aims to establish (iii) the relationship between choices (i) and (ii), i.e. whether there is a correlation between the use of the accusative vs. dative in the active construction and the use of the middle-*se* construction.

The study is rooted in the concept of construction as a conventional association of meaning and form (CxG), motivated by communicative goals. This approach posits that grammatical forms emerge from the discourse context to serve the expressive needs of speakers.

Accordingly, our analysis of the syntactic behaviour of psychological verbs is based on the use of the constructions registered in corpora, paying particular attention to the frequency and relative distribution of the variable features that characterise them. The importance of the frequency of use derives from the assumption that the most frequent combinations of linguistic properties reflect the communicatively most efficient meanings (cf. Levshina 2023). In the rest of the paper, we first describe the data sources for the study -the ADESSE database and CORPES XXI- and the method used to extract the data sets for the statistical tests (section 2). Section 3 then presents the results of the logistic regression analyses. The factors influencing the experiencer's syntactic coding (subject vs. object) are examined, focusing on the contrast between Psych-I and Psych-II verbs, on the one hand, and on the diathesis alternation between active and middle-*se* constructions with Psych-IIa verbs, on the other. Next, section 4 deals with the accusative/dative alternation with Psych-IIa verbs. Section 5 examines the correlation between the frequency of accusative/dative case marking and the frequency of middle-*se* constructions. Finally, section 5 discusses and sums up the conclusions of the study.

2. Data and methodology

For this study, we have used two sources of data: ADESSE and CORPES.

ADESSE (<http://adesse.uvigo.es/>) is a database with detailed lexical, semantic, and syntactic information about the argument structure of 159.000 clauses of a 1.45-million-word corpus (ARTHUS) of contemporary Spanish texts published between 1980 and 1990. In the ADESSE database, clauses are analysed for many syntactic and semantic properties, and the around 3500 verb lemmas ($\pm$ 4.000 lexical entries) are classified into a detailed conceptual typology of processes.

CORPES (<https://www.rae.es/corpes/>) is a lemmatised and POS tagged reference corpus of Spanish with a size of 395 million words (as of version 1.0), which consists of a balanced sample of Spanish texts of the XXI century.

From ADESSE we have selected all verbs and clauses classified as “*Sensación*” (i.e., Emotion/Psych). This amounts to a dataset of 5,845 clauses, and 265 different verbs. The dataset includes both Psych-I verbs (1,608 clauses, 56 verbs) and Psych-II verbs (4,237 clauses, 224 verbs), and detailed semantic and syntactic properties of the clause (Voice, TAM, ...) and its arguments (Semantic Role, Syntactic Function, Case, Gender, Number, Animacy, etc.). Figure 1 summarize the constructions of the ADESSE data.

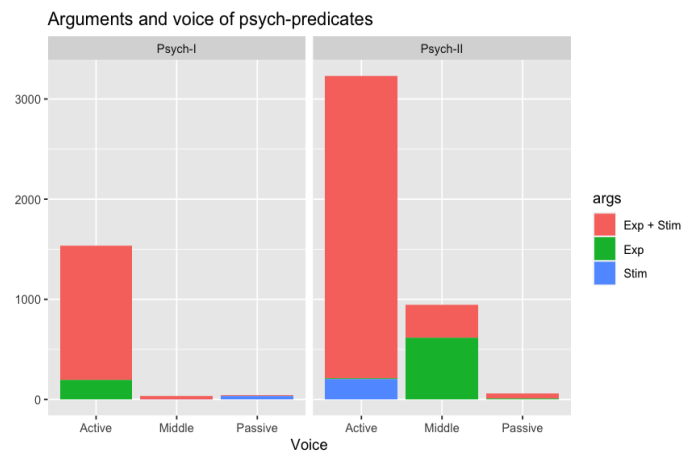


Figure 1. Syntactic constructions, voice, and arguments of Psych-verbs in ADESSE data

As for CORPES, we have used its search engine to extract the inventory of grammatical forms consisting of a psych verb plus a personal pronoun (which corresponds to the Experiencer) immediately preceding or following the verb form (for example: *lo alegra, se alegró, alegrar-se, me he alegrado, alegra-te*, etc.) and the frequency of each combination. We have performed independent searches for 50 Psych-IIa verbs known to be highly frequent. As Psych-IIa verbs we understand verbs that allow the accusative case for Experiencer-Object (as in *lo alegra* ‘it makes him happy’) and Middle voice with the Experiencer-Subject (as in *se alegró* ‘he was happy’). Here is the list of the 50 selected psychological verbs:

aburrir 'bore', *agobiar* 'overwhelm', *alarmar* 'alarm', *alegrar* 'gladden', *animar* 'animate', *apaciguar* 'appease', *apasionar* 'be passionate about', *aplacar* 'calm down', *asombrar* 'amaze', *asustar* 'frighten', *atormentar* 'torment', *aturdir* 'stun', *avergonzar* 'embarrass', *calmar* 'calm', *cansar* 'tire out', *complacer* 'please', *conmover* 'move', *consolar* 'comfort', *decepcionar* 'disappoint', *desconcertar* 'bewilder', *disgustar* 'displease', *divertir* 'amuse', *enfadar* 'anger', *enojar* 'anger', *enorgullecer* 'make/be proud', *entretener* 'entertain', *entristecer* 'sadden', *entusiasmar* 'delight', *escandalizar* 'scandalize', *estremecer* 'shake/shudder', *excitar* 'excite', *extrañar* 'miss', *fastidiar* 'annoy', *fatigar* 'tire', *horrorizar* 'horrify', *incomodar* 'disturb', *indignar* 'outrage', *inquietar* 'disquiet', *irritar* 'irritate', *joder* 'piss off', *molestar* 'bother', *obsesionar* 'obsess', *ofender* 'offend', *preocupar* 'worry', *reanimar* 'reanimate', *relajar* 'relax', *satisfacer* 'satisfy', *sobresaltar* 'startle', *sorprender* 'surprise', *tranquilizar* 'reassure'.

Each verb was searched for the subcorpora of Spain or American origin, which makes 100 searches (50 x 2). Once the results were filtered and cleaned, we got a dataset of 12,034 records (Verb forms + Pronoun forms), corresponding to 169,338 total cases (clauses), from which 104,548 are from the American subcorpus and 64,790 are from Spain.

As CORPES does not provide tags with the search results, an automatic tagging was performed to get morphosyntactic properties of verb forms (TAM, person, number) and pronoun forms (person, number, case), followed by an automatic assignment of a syntactic function (Subj / Obj) to the Experiencer, based on the agreement or lack of agreement in person and number between verb form and pronominal form.

The final datasets contain the variables and levels listed in Table 1. The lack of semantic annotation in CORPES determines the selection of independent variables. Compared to ADESSE, in CORPES it is not possible to differentiate between animate and inanimate Stimuli. On the other hand, in ADESSE we cannot get enough data on specific verbs, and they are aggregated into verb types; whereas in CORPES there are many data about all of the 50 selected verbs.

Table 1. Factors and levels of ADESSE and CORPES datasets¹

FACTOR	LEVELS	Dataset
ZONE	Es [Spain] vs. Am [America]	ADESSE, CORPES
TEXTGENRE	Essay vs. Narrative vs. Press vs. Spoken vs. Theater	ADESSE
(VERB) TYPE	Psych-I vs. Psych-II	ADESSE
VERB	[50 selected Psych-IIa verbs]	CORPES
ASPECT	Perf vs. Imperf ²	ADESSE, CORPES
EXP_FSYNT	Subj vs. Obj	ADESSE, CORPES
EXP_PERSON	1 vs. 2 vs. 3	ADESSE, CORPES
STIM_PERSON	1 vs. 2 vs. 3	ADESSE, CORPES
STIM_ANIM	an [animate] vs. in [inanimate] vs. inc [clause]	ADESSE

Over these datasets we have performed multivariate logistic regression analyses. In each step we have filtered the relevant data from the datasets and calculated a model that included all the potentially relevant predictors. We have used the glm() function of the base R software (R Core Team 2021) and the lrm() function of the RMS package (Harrell 2020), to determine in each model which factors were statistically significant, their effect size and the relative importance of each predictor.

¹ Abbreviations: EXP=Experiencer, STIM= Stimulus, FSYNT=Syntactic Function, ANIM=Animacy

² Perf= simple past perfect and all compound perfects. Imp= all simple tenses except simple past perfect

3. Results: Experiencer-Object vs Experiencer-Subject

The syntactic function of the Experiencer depends on the verb type and the voice. In the active unmarked voice of Psych-I verbs (as *temer* ‘fear’ or *amar* ‘love’) the Experiencer is the Subject; whereas with Psych-II verbs (as *asustar* ‘frighten’ or *alegrar* ‘worry’) the Experiencer is the Object in the active voice clauses and the subject in the middle-*se* clauses. We want to explore which independent factors may influence the choice of the syntactic function of the Experiencer and whether those factors are the same when we choose between the two subclasses of psych verbs and when we choose the voice of Psych-IIa verbs.

3.1. Experiencer Subject (Psych-I) vs Experiencer Object (Psych-II) in Active voice

EXP=SUBJ (*él teme algo*) vs EXP=OBJ (*algo le/lo preocupa*).

This choice between Experiencer Subject construction (Psych-I) and Experiencer Object construction (Psych-II) has been analysed by Vázquez-Rozas and Miglio (2016), and they have found that Experiencer person and number, Stimulus animacy, and Text Genre are significant for the selection of the construction. We have now performed a new regression analysis adding the variables Zone (with two levels: Spain vs. America) and Aspect (with two levels: Perf and Imperf) and we have confirmed that both the person of the Experiencer and the animacy of the Stimulus are significant predictors for the selection of the construction: 1st and 2nd person Experiencers tend to be cast as Object (Psych-II), while 3rd person Experiencers prefer the Subject function (Psych-I). Inanimate Stimuli, and particularly propositional ones, correlate significantly with Exp-Obj (Psych-II verbs), while animate stimuli are mainly found in Exp-Subj constructions (Psych-I verbs). However, the newly introduced variables, Zone and Aspect, are not significant predictors. The most important predictor is the Text Genre: the Spoken subcorpus prefer Psych-II verbs, and the Press subcorpus prefer the Psych-I verbs. The model yields an acceptable discrimination index $C=0.7$

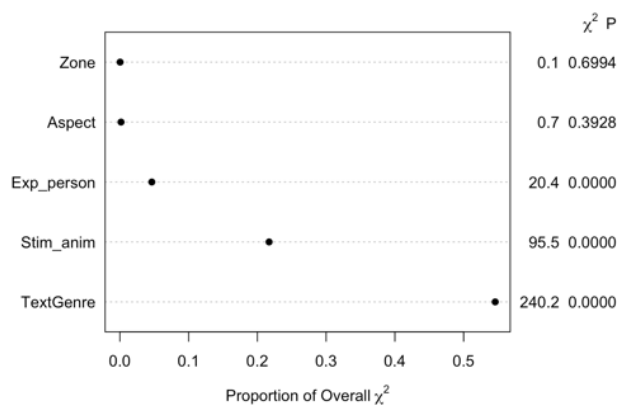


Figure 2. Psych-I (Exp=Subj) vs Psych-II (Exp=Obj). Relative importance of predictors (ADESSE data [N=4353 clauses of 218 psych verbs in active voice])

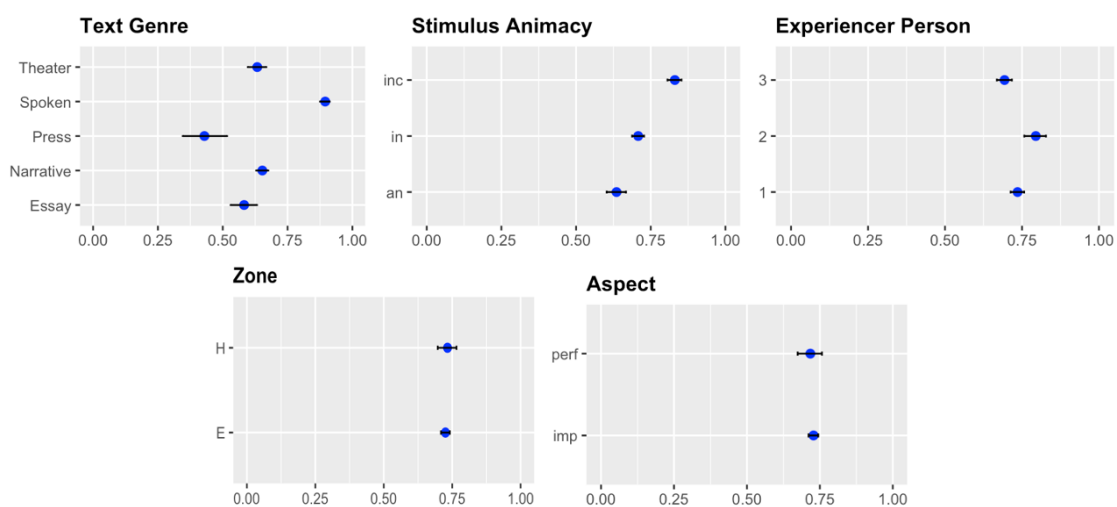


Figure 3. Probabilities of Psych-II (Exp=Obj) vs. Psych-I (Exp=Subj) as predicted by the model (ADESSE data [N=4353 clauses of 218 psych verbs in active voice])

3.2. Psych-II: Active vs Middle voice in ADESSE data

EXP=OBJ (*algo le/lo preocupa*) vs EXP=SUBJ (*él se preocupa por algo*)

The selection of the Experiencer as Subject I is what characterizes the middle voice construction with class II verbs, which tends to include the Experiencer as the only argument. Obviously, if there is not a Stimulus argument, the only option is to use the EXP-SUBJ middle construction, which does not require the expression of the Stimulus. However, the Stimulus is expressed in a third of the occurrences of the middle construction (see the right part of Figure 1).

We have performed a regression analysis for the choice between active (Exp=Obj) [3018 cases] and middle (Exp=Subj) [947 cases] constructions [total 3965 cases of 265 different verbs]. The model includes the variables Text Genre, Experiencer person, Zone and Aspect; but not any property of the Stimulus because in most cases it is null. We have found the

variables Aspect, Person of the Experiencer, and Text Genre as the most relevant predictors of the voice constructions within Psych-II verbs, while Zone is not a significant factor. The model yields an acceptable discrimination index $C=0.713$

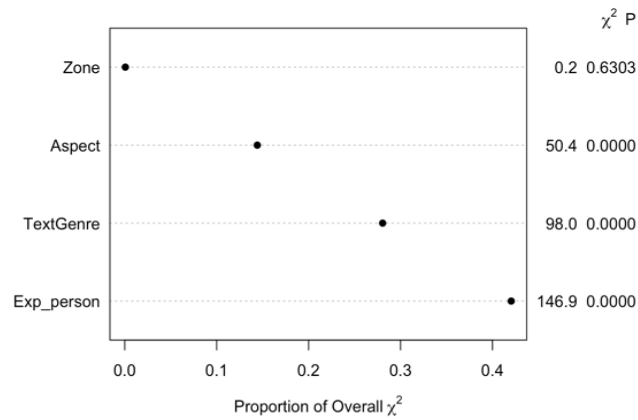


Figure 4. <Active: Stim=Subj Exp=Obj> vs <Middle: Exp=Subj (Stim=Obl)>:
Relative importance of predictors (Psych-II verbs in ADESSE dataset)

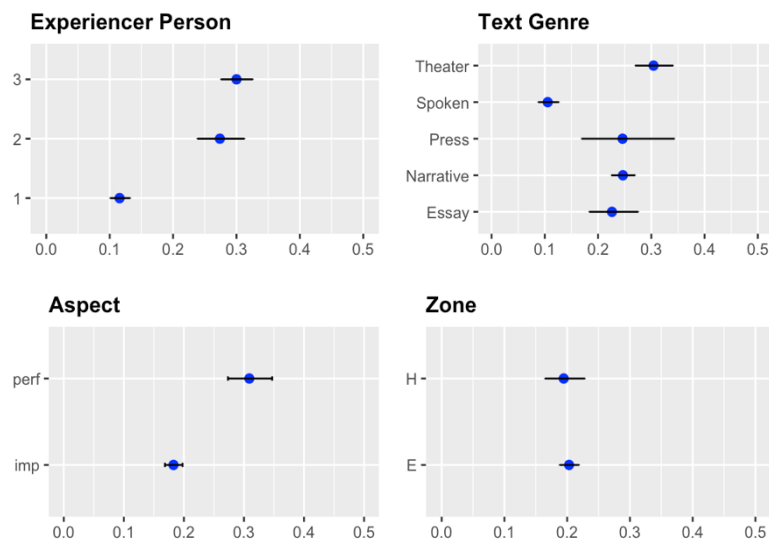


Figure 5. <Active: Stim=Subj Exp=Obj> vs <Middle: Exp=Subj (Stim=Obl)>:
Predicted proportions of Middle construction (Psych-II verbs in ADESSE dataset)

The results, as can be seen in Figure 5, show a higher proportion of Middle construction in non-1st person, non-spoken text genre, and perfect aspect.

3.3. Analysis and results of CORPES data: Exp=Obj (Active) vs Exp=Subj (Middle)

For this research, we have data from CORPES on the combination of Verb + Pronoun with 50 Psych-II verbs, from which we have been able to obtain several morphosyntactic characteristics derived from the nominal and verbal morphology. However, the data extracted from CORPES do not allow us to evaluate the TextGenre variable, because it was not easy (although it would not have been impossible) to obtain it with the search system; but the Zone

variable is because we have searched separately for data from Spain and America. We can also analyse, as in the models used with the ADESSE dataset, the variables Aspect and Experiencer person; but for the Stimulus the only data we can obtain is the verbal number and person when it functions as a subject (in the active voice). In return, we have enough data on each verb to introduce it as one of the possible relevant factors. The corresponding model shows, in fact, that this is the main determining factor in the choice between Active (Exp=Obj) and Middle (Exp=Subj). From the rest of the factors, we obtain with these 50 verbs results very similar to those we had obtained in ADESSE with a larger number of verbs and a much smaller number of exemplars.

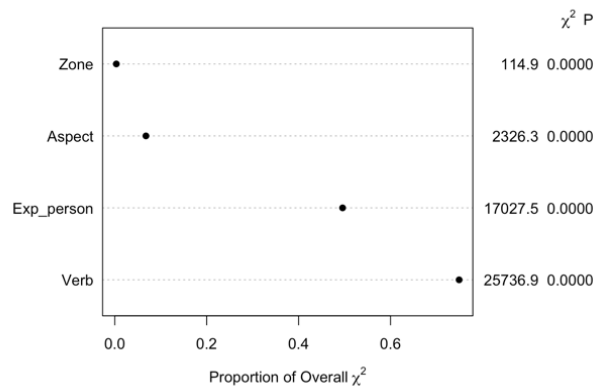


Figure 6. Exp=Subj (Middle Voice) vs Exp=Obj (Active Voice). Relative importance of predictors. (CORPES dataset: 169,338 clauses of 50 psych-II verbs)

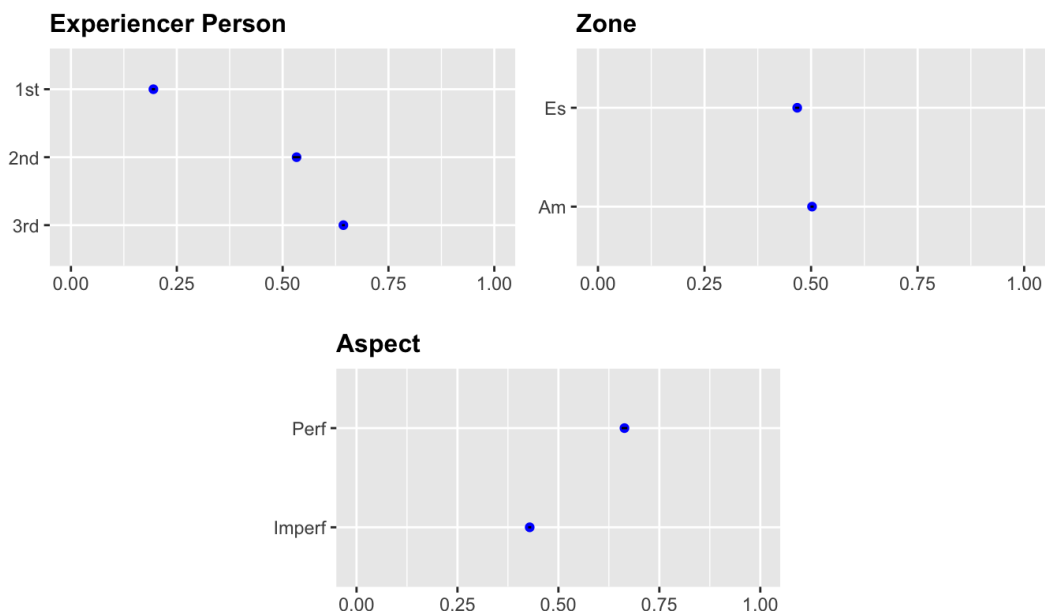


Figure 7. Probabilities of Exp=Subj as predicted by the model (CORPES data)

The probability of Middle Voice (Exp=Subj) increases with the Perfect Aspect and non-1st Person Experiencers, while the Imperfect aspect and 1st Person Experiencer favor Active

Voice (Exp=Obj). The light preference for Middle Voice in American texts is significant because there are many data, but the effect size is very small. The results are quite like those obtained from ADESSE data. But now the number of tokens provided by CORPES for each verb gives strong statistical support to the differences between verbs shown in Figure 8.

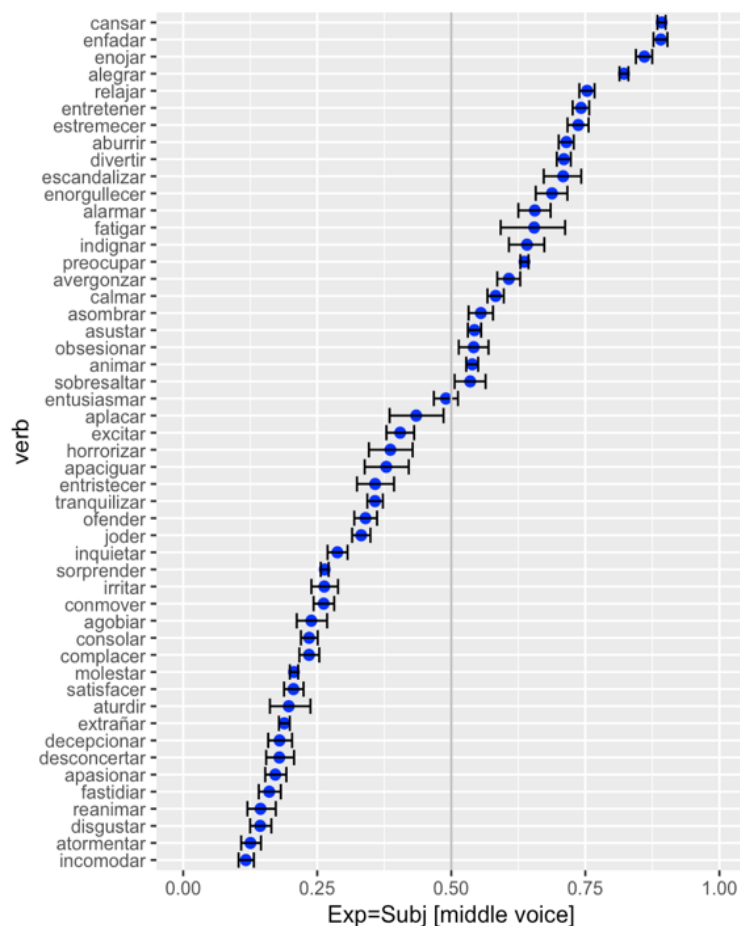


Figure 8. Predicted proportions of Exp=Subj (& Middle Voice) for each verb (CORPES data)

The verb rank in Figure 8 is likely related to the spontaneity of the event, with verbs at the top end (like *cansarse* ‘get tired’) referring to events that occur more spontaneously as an internal process, and verbs at the bottom (like *incomodar* ‘disturb’) requiring the involvement of a participant external to the Experiencer. Some (near-)synonyms, like *enfada* and *enojar* ‘anger’, show a similar behaviour in their preference for the middle construction. However, other near-synonyms behave differently: for example, *fatigar* is associated with the middle construction less than *cansar*, although both are situated near the top; verbs related to the meaning ‘bother’ (*joder*, *irritar*, *molestar*, *fastidiar*) also behave differently although they are situated rather towards the bottom, and verbs whose meaning is ‘calm down’ or something similar (*relajar*, *calmar*, *apacar*, *apaciguar*, *tranquilizar*) are distributed along different points of the scale. It seems that there is always something lexically idiosyncratic.

3.4 The syntactic function of the Experiencer: summary of results

The choice of syntactic function of the Experiencer (Subject or Object) is associated with verb class (Psych-I vs. Psych-II) or with grammatical voice within Psych-II verbs (Active vs. Middle pronominal construction). After analysing the effect of several independent variables on this choice, we summarize here the main results:

There are significant differences in use across Text Genres: the Spoken subcorpus of ADESSE/ARTHUS shows a higher probability of Psych-II verbs (vs Psych-I) and Active Voice (vs. Middle) within Psych-II verbs, i.e. there is a consistent preference for constructions with Experiencer as Object in spoken registers.

On the contrary, we have not found significant differences between Zones (Spain vs. America) neither in the choice of verb types nor in the preference for any voice.

Besides Text Genre, Experiencer Person is one of the most relevant predictors of the choice constructions: 3rd person Experiencers are more associated with EXP-SUBJ constructions (either Psych-I, as in *temen* ‘they fear’, or middle voice Psych-II, as in *se asustan* ‘they got scared’); while 1st person Experiencers are more associated with EXP-OBJ constructions (Active voice Psych-II, as in *me asustan* ‘they scare me’). The prominence of the speaker in psychological events is determined by empathy and topicality features rather than by the animacy hierarchy (more related to agency).

Aspect is not significant for the choice between Psych-I and Psych-II; but it is significant within class II for the voice alternation: the middle EXP-SUBJ is more likely with the perfect aspect, and active EXP-OBJ with the imperfect. Frequency of perfect aspect is correlated with telic Aktionsarten (Becker & Malchukov 2022), so that we can infer that the middle voice tends to construe the event as an effective change of state. On the contrary, in the active voice both Psych-I and Psych-II tend to construe the event as stative, without significant aspectual differences between the two types of verbs, despite the differences in syntactic patterning.

Stimulus animacy is a significant predictor of the contrast between Psych-I and Psych-II. However, this variable has not been included in the statistical models for the class II alternatives, because of the sparseness of ADESSE data, and the absence of this kind of information in CORPES data.

Finally, the lexical verb is a very significant factor for the choice of the syntactic constructions. The syntactic function of each participant in a psychological process is at large lexically determined. This is obvious for the alternation between Psych-I and Psych-II

constructions, where we have two (almost) disjoint classes. Within the Psych-II class, some verbs allow voice alternation and some others do not. And within those verbs that allow the voice alternation, we have found in CORPES data that there is a continuous cline from verbs that show a preference for the use of the middle voice EXP-SUBJ construction (like in *cansarse*) to verbs that show a preference for the use of the active voice EXP-OBJ construction (like in *incomodarlo*). The possibility of middle voice is related to transitivity, and the relative probability of using middle voice construction may be also related to transitivity. We will explore this possibility in the following sections.

4. Case alternation with Exp-Obj verbs: Accusative vs Dative

The verbs that allow the middle voice construction also alternate in the active voice for the Experiencer as Object between the Accusative (*lo preocupa*) and the Dative (*le preocupa*) case. This alternation is limited to Psych-IIa verbs and can be seen only in 3rd person pronominal clitics, as 1st and 2nd person clitics (*me preocupa*, *te preocupa*) do not manifest this case distinction. The Experiencer-Object is most frequently 1st person, as it is shown in Figure 9. Nevertheless, in the CORPES dataset we have got 38,072 valid cases to explore this alternation from which 19,779 of Accusative and 18,293 of Dative.

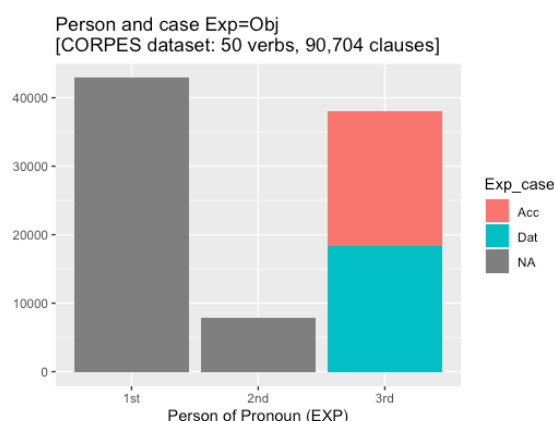


Figure 9. Person and case of Experiencer pronominal clitics Object (CORPES dataset)

In the statistical model we have introduced as independent variables the same factors as in the analysis of the voice alternation, with the exclusion of Experiencer person (not relevant because case alternation is limited to 3rd person) and the inclusion of Stimulus person, which is recoverable from the agreement with subject in verb morphology. Figure 10 and Figure 11 illustrate the main results.

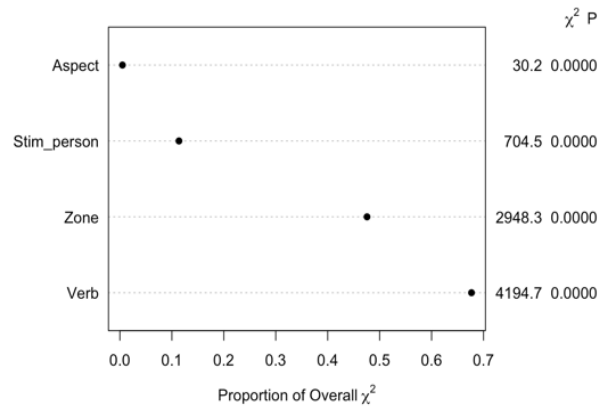


Figure 10. Accusative (*lo / la*) vs Dative (*le*). Relative importance of predictors. (CORPES dataset)

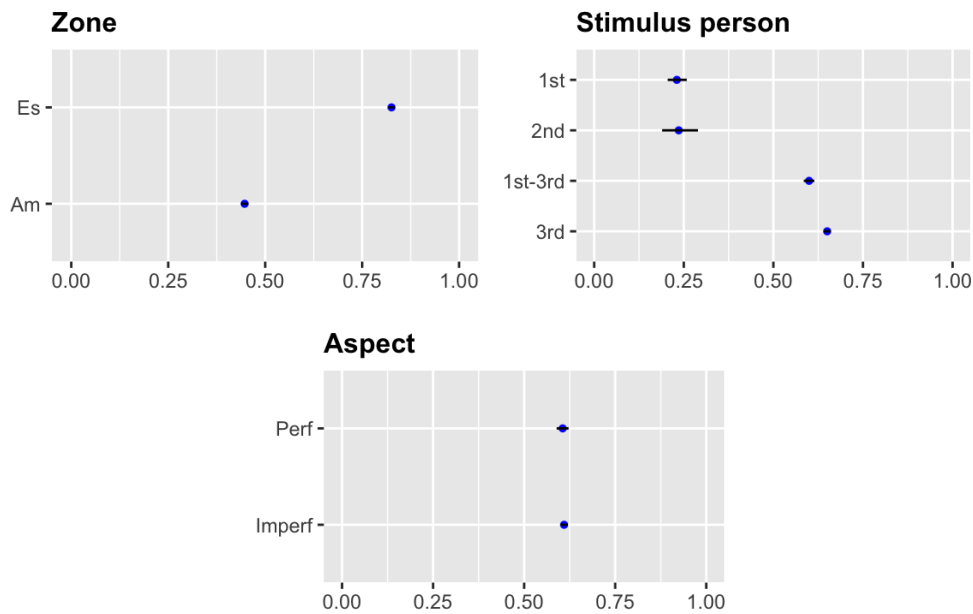


Figure 11. Predicted probabilities of Exp=DAT with selected predictors (CORPES data)

Although all factors appear to be significant (given the large amount of data), Aspect is the less relevant with a slight preference for Accusative with Perfect tenses, and Dative with Imperfect tenses.

Verb person indexes the Subject argument and is related to Stimulus animacy: 1st and 2nd person forms always have human referents while 3rd person forms can refer to both animate and inanimate entities. Therefore, 1st and 2nd person Subjects correlate with Accusative case, as in *la agobio* [*yo*] ‘I pester her’, while 3rd person forms show a higher proportion of Datives, as in *le agobia* [*eso/ella*]. Some forms of imperfect and conditional tenses are ambiguous between 1st and 3rd person (*le agobiaba* [*yo / ella*]), and they also show a higher proportion of Datives, which suggest that these forms correspond to 3rd person referents in most cases.

Regarding the variable Zone, American texts correlate with Accusative case while texts from Spain prefer Dative case. This is clearly a consequence of the prevalence of *leísmo* in Peninsular Spanish.

Finally, the verb lemma is the strongest predictor of the Object case, which seems to be lexically determined, with a cline going from verbs like *preocupar* ‘worry’ and *disgustar* ‘upset’ with the highest proportion of Dative (*le preocupa* ‘(s)he/it worries him/her’) to verbs like *fatigar* ‘tire’ and *reanimar* ‘revive, encourage’ with the highest proportion of Accusative (*lo reanima* ‘(s)he/it revives him’), as shown in Figure 12. Each verb shows a homogeneous behaviour in its preference for Accusative or Dative regardless of the other independent variables.

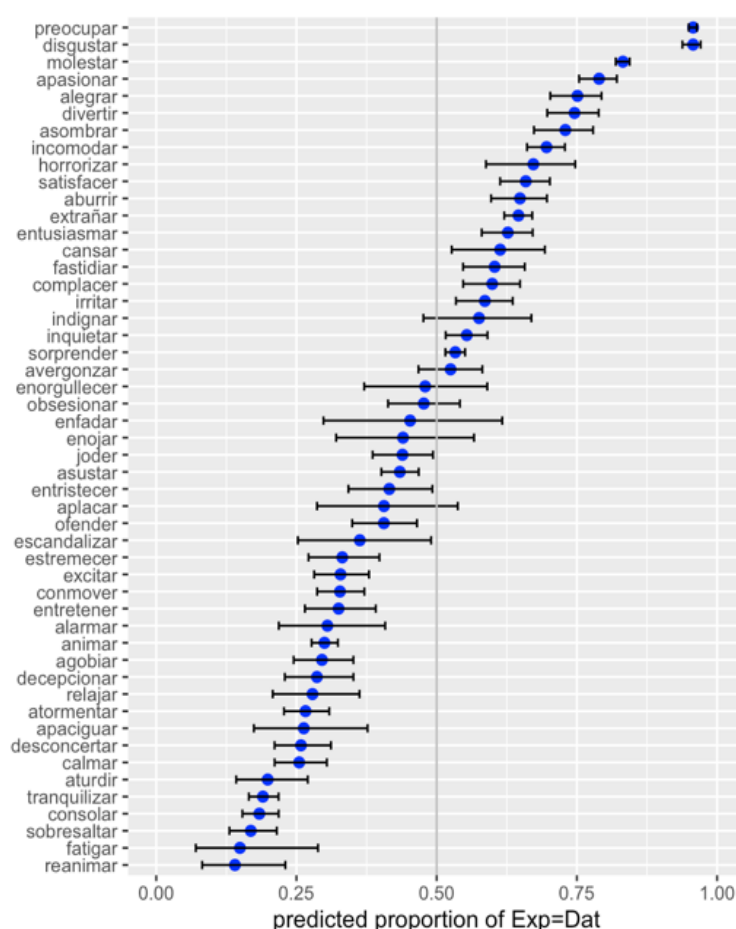


Figure 12. Predicted probabilities of Exp=DAT (vs Acc) for each verb (CORPES data)

The preference for Accusative or Dative case with psych verbs has been related to transitivity, and they can be seen as differences in relative 'kinesis', 'agency', and/or 'object affectedness' (Whitley 1995: 581). However, the lexical meanings do not show clear tendencies and there must be something idiosyncratic in lexical preferences. Some synonyms, like *enojar* and *enfadar*, behave similarly and we can find them together in the middle of the

scales, whereas other synonyms like *cansar* and *enojar* ‘tire’ are clearly separated: *cansar* is situated near the top, associated with a preference for the dative, and *fatigar* is situated almost at the bottom, associated with a preference for the accusative.

5. Case alternation and diathesis alternation

Given that psych verbs class IIb always take dative case in the EXP-OBJ construction and consistently reject the EXP-SUB middle construction (“anticausative”), whereas Psych verbs class IIa alternate accusative and dative case in their EXP-OBJ construction and alternate EXP-OBJ and EXP-SUB middle construction (show ‘anticausative’ alternation), then it can be hypothesized that the frequency of use of ACC case for the object of Psych-IIa verbs correlates with the relative frequency of their use of the EXP-SUB middle construction. Is there a relation between case alternation and voice alternation? If there is, which is the direction of the correlation? The Figure 13 is a scatterplot whose axes are the data of Figure 8 and Figure 12, i.e., for each of the 50 verbs of Corpes data set, the predicted proportions of Exp=Subj (& Middle Voice) on the horizontal axis, and the predicted proportions of Exp=Acc (vs Exp=Dat) on the vertical axis.

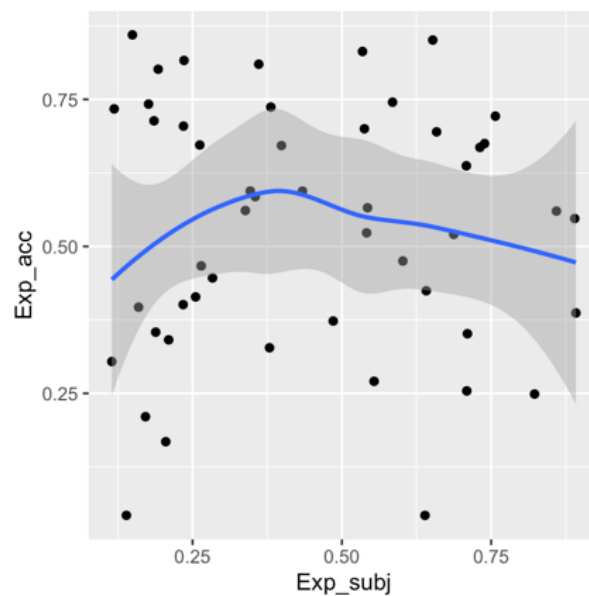


Figure 13. Proportions of Exp=Acc and Exp=Subj for each verb, as predicted by regression models. (Each point represents a verb) [Pearson’s correlation coefficient $\rho=0.005$, p-value=0.975]

The smooth line shows that there is no correlation, neither positive nor negative, between the preference for Accusative or Dative in the active voice and the higher or lower use of the Middle Voice. A correlation test confirms that there is absolutely no correlation between those two variables ($\rho=0.005$, p-value=0.975).

Why is the relationship between EXP-OBJ_{ACC} and EXP-SUBJ (“anticausative”) dismissed? It is due to the functional differences between the two constructions. Both constructions share a correlation with perfect aspect, although it is less significant for ACC case. They contrast sharply in the Stimulus’ features: EXP-OBJ_{ACC} correlate with 1P & 2P stimuli. EXP-SUBJ either do not mention the Stimulus at all or, when it is expressed, it is inanimate.

6. Discussion and conclusion

The main difference between the active <Stim=Subj - Exp=Obj> construction and the middle <Exp=Subj> alternative lies in the presence vs absence of the Stimulus as a core argument of the clause. Whereas in active Exp-Obj clauses the Stimulus is encoded as a subject, in middle Exp-Subj clauses it is either not expressed at all or is downgraded to an oblique.

This syntactic difference relates to the communicative function of each type of construction. The active construction, which gives prominence to the Stimulus (the subject), is associated with 1sg Experiencers. The 1st person singular Experiencer introceptively perceives his inner feelings and emotions and is aware of the Stimulus that triggers them, which is not directly accessible to outsiders. His/her awareness of the entity or situation that brings about the emotion (the Stimulus) leads 1st person speakers to include it as a core participant in the event, and therefore to favour the two-argument active pattern <Stim=Subj - Exp=Obj> instead of the middle <Exp=Subj>, which only allows for a single core participant, the Experiencer.

In contrast, 3rd person Experiencers prefer the middle Exp-Subj constructions, which do not require the Stimulus to be specified, although causal-related information can be expressed either as an oblique (8-9), as an adjunct clause (10), or as another type of construction (11).

- (8) *se extrañó de aquella brusca aparición de Jano con la ropa llena de hojas secas.* [CAR:086.33]
‘She was surprised by that sudden appearance of Janus with his clothes full of dry leaves.’
- (9) *Arancha se enfadó en alguna bola dudosa* [3VO:045-1.0-01]
‘Arancha got angry at some dubious ball.’
- (10) *Ella se alegra al verle* [SON:205.01]
‘She is happy to see him.’
- (11) *la muchacha levanta al niño y lo estrecha en un vivo gesto tan instintivamente maternal que el viejo se conmueve.* [SON:105.31]
‘The girl picks up the child and hugs him in a vivid gesture so instinctively maternal that the old man is moved.’

The ‘cause phrases’ (so called by Babcock 1970: 41) underlined in (8)- (11) refer to observable events that precede or accompany the emotions attributed to third-person Experiencers. This external point of view, which contextualises the reporting of others’ emotions, is consistent with Brinton’s (1980: 370-371) ‘outer perspective’ as opposed to an ‘inner perspective’, and with Wiebe’s (1994) ‘private-state action’ as opposed to a ‘private-state report’. Wiebe points out that, in third-person narrative, the perceptions and feelings of the characters can either be expressed through ‘private-state reports’ (*She was unhappy*), as is common in first-person narrative (*I am happy*), or inferred from ‘private-state actions’: “The sentence *She frowned* narrates a private-state action from which unhappiness or displeasure can be inferred, but does not directly report the character’s private state” (Wiebe 2002: 264). As examples (8)- (11) show, events causally related to the generation of the emotion expressed by the middle Exp=Subj clauses usually denote dynamic and telic situations. On the contrary, the stimuli of the active constructions are propositional entities without spatio-temporal anchoring, as can be observed when comparing (12) and (13)

- (12) *Ella se alegra al verle* (‘ella se alegra en el momento de verle / cuando le ve’)
‘She is happy to see him’ (i.e., she is happy when she sees him)
- (13) *A ella la/le alegra verle* (‘a ella la/le alegra el hecho de / *el momento de verle’)
‘She is happy to see him’ (i.e. The fact of seeing him makes her happy)

This difference in the conceptualisation of the Stimulus is also related to the aspectual interpretation of clauses with Psych-IIa predicates: the middle Exp=Subj construction is expected to be associated with the perfective aspect, which is also consistent with its 3rd person preference. As previously mentioned, the description of psychological states of others generally requires observing the person’s reactions and visible physical changes. This implies a “narrative” distance between the event and its linguistic description, which favours the use of the perfective aspect.

Besides tense and person/animacy variables, which have been shown to correlate with both diathesis and case alternation, the statistical analysis underlines the importance of the verb as a key factor in the choice of construction. However, the significance of the correlation of the verb lemma with diathesis alternation is independent of that with case alternation, suggesting that the basis of the verb-construction association is idiosyncratic in nature. Figures 12 and 13 confirm such a conclusion and exhibit the different syntactic behaviour of near-synonymous verb pairs such as *cansar* and *fatigar*, or *aplacar* and *relajar*.

Finally, it should be noted that the middle EXP-SUBJ construction is seen as a transitivity reducing mechanism, since it omits one of the two central participants, the Stimulus. Therefore, active EXP-OBJ clauses are more transitive than middle ones in terms of the number of participants (parameter no. 1 of Hopper & Thompson 1980). On the other hand, middle clauses with Psych-II verbs are associated with the perfect aspect, which characterises them as more transitive than active clauses, which are correlated with the imperfect aspect. Moreover, the subject experiencer of the middle construction shows characteristics related to the agency (control, volition), for example being compatible with the imperative on occasions, both in the negative *-no te enfades / preocupes / disgustes / fatigues / avergüences* ‘do not be angry / worry / be upset / get tired / be ashamed’ - and in affirmative sentences: *preocupaos de los vuestros* ‘worry about yours’; *Cálmese, señorita* ‘calm down, miss’; *Ánimate, mamá* ‘cheer up, mom’; *Alégrate* ‘get happy’ [ADESSE examples].

The association of middle-*se* clauses with the perfective aspect and third person subjects specialises them as a syntactic device for foregrounding psychological events attributed to characters in narratives. Note that, according to Hopper & Thompson’s (1980) transitivity hypothesis, the pragmatic function of high transitive clauses in discourse is precisely that of foregrounding information, in contrast to the low transitive clauses, which tend to be associated with background information.

The whole picture shows that the frequency tendencies are a result of the interplay of several different lexical, syntactical, and communicative factors, which can not be reduced to a single principle.

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